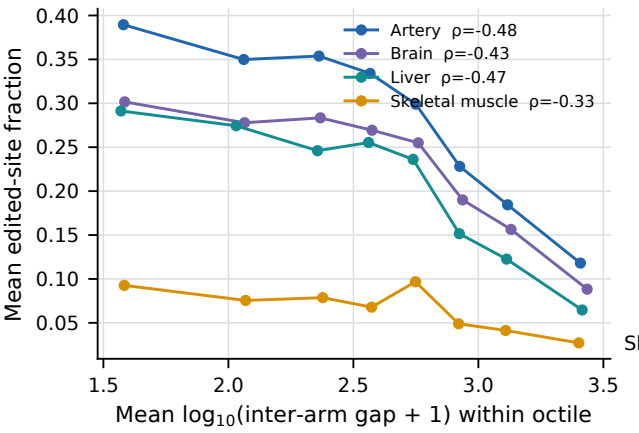
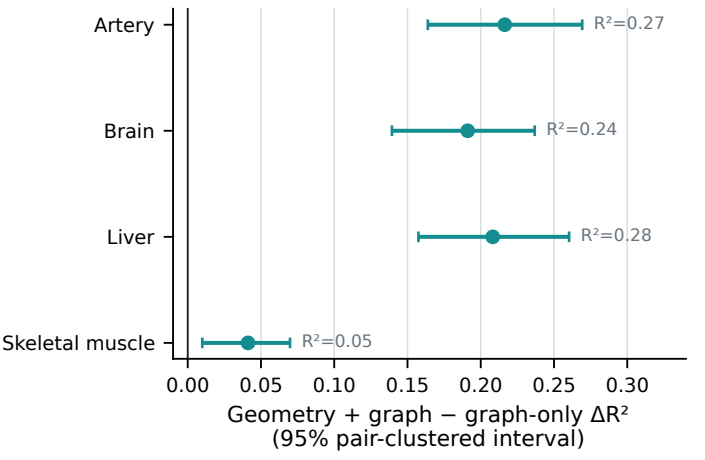
A

Evidence map: what the benchmark can and cannot currently resolve

Scale	Tested formulation	Status	Interpretation
Substrate-aware learning	Pair-first expanded supervision	Selected	Corrects label use and pair weighting
Target selectivity	Aligned target + helix-partner view	Selected	Survives shifted-window control
Integration	Fixed whole-duplex + local-augmented logits	Selected	Complementary same-fold predictions
Substrate propensity	Pair geometry / editing burden	Supported*	Editing burden is learnable; latent state unresolved
Supervision	Continuous + ordinal auxiliary labels	Limited	Ranking benefit; no useful F1 gain
Target selectivity	Focal node / contact geometry	Limited	Points toward aligned local view
Cross-tissue	Shared four-tissue training	Limited	Transfer below promotion criterion
Substrate propensity	Hierarchical engagement gate	Limited	Small ranking gain; unstable classification
Biological context	Bulk H3K36me3 overlap	Limited	Context-specific AUPRC only
Biological context	RBP binding × bulk expression	Not supported	No reproducible residual gain
Biological context	Transcript / repeat annotation panel	Not supported	No robust site-level increment
Representation	Frozen RNA-language embeddings	Not supported	No consistent incremental utility
Representation	Substrate-wide all-A decoder	Not supported	Stopped at prespecified futility gate
Capacity	Larger / deeper graph networks	Not supported	Capacity alone did not generalize
Integration	Unified end-to-end architecture	Not supported	Inferior to frozen ensemble
Cellular state	Matched donor/cell multimodality	Deferred	Required measurements unavailable
RNA state	Matched in vivo structure / isoforms	Deferred	Required measurements unavailable
Capacity	Graph Transformer / GATv2 expansion	Not pursued	Superseded by controlled lower-capacity evidence

Status reflects prespecified same-fold comparisons.
*Reproducible observation not yet actionable for site prediction; deferred hypotheses were not tested.

B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